

CloudFront Custom Domain and HTTPS with an S3 Origin

To Begin with the Lab

Summary of the Lab

In this lab, an **Amazon S3** static website was connected to **Amazon CloudFront** and exposed through a custom branded domain with HTTPS. A certificate was requested in **AWS Certificate Manager**, validated through **Route 53** DNS records, and issued in **US East (N. Virginia)** because CloudFront requires the certificate to be there. A CloudFront distribution was then created using the S3 website endpoint as the origin, HTTP-to-HTTPS redirection was enabled, and the custom SSL certificate was attached. After that, a Route 53 alias record was created to point the custom domain to the CloudFront distribution. Finally, the website was verified through the branded HTTPS URL, while direct S3 access was still observed as a security gap to be handled in the next lab.

Understanding CloudFront with a Custom Domain and HTTPS

A custom domain makes a website look trusted and professional instead of exposing the default CloudFront URL. **HTTPS** exists so the browser can encrypt traffic and prove the site is safe to open, which is important for business websites. **CloudFront** sits in front of the origin so the site can be delivered securely while still using a branded address like cloudfreeks.com. The combination of **Route 53**, **AWS Certificate Manager**, and **CloudFront** exists to give users a secure, validated, and easy-to-remember web address.

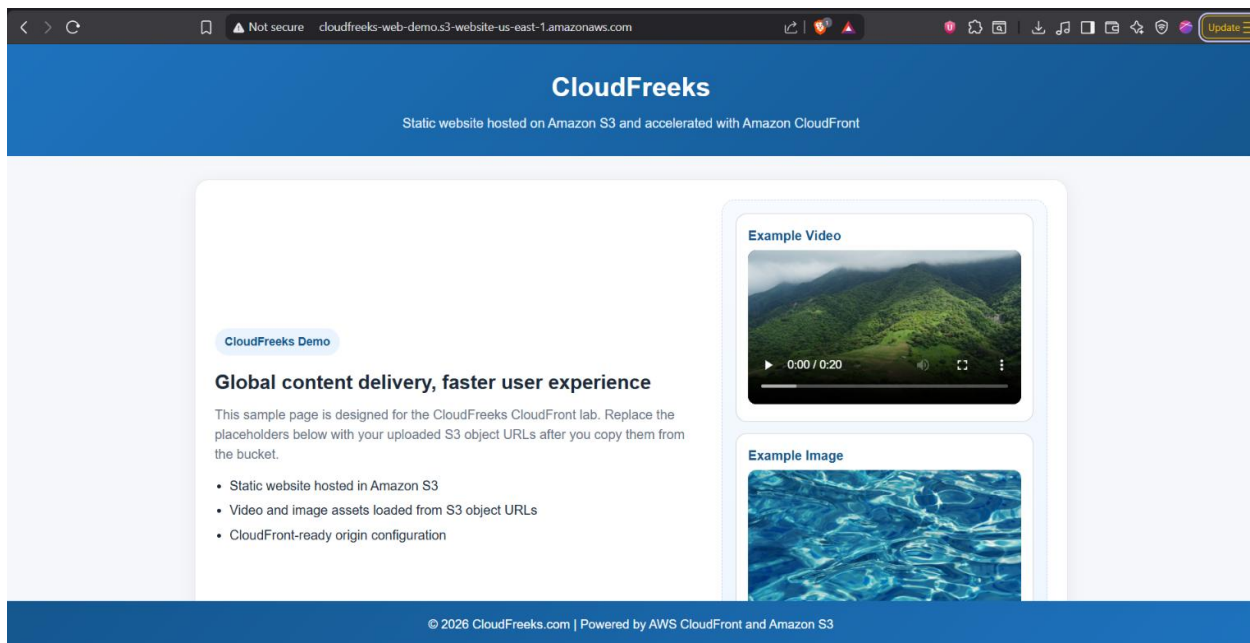
Example:

A visitor opens cloudfreeks.com and sees a secure lock icon in the browser instead of a generic cloudfront.net address. That makes the site feel trustworthy and shows that the connection is encrypted before any content is loaded.

Lab Steps

Step 1 — Prepare the S3 Static Website

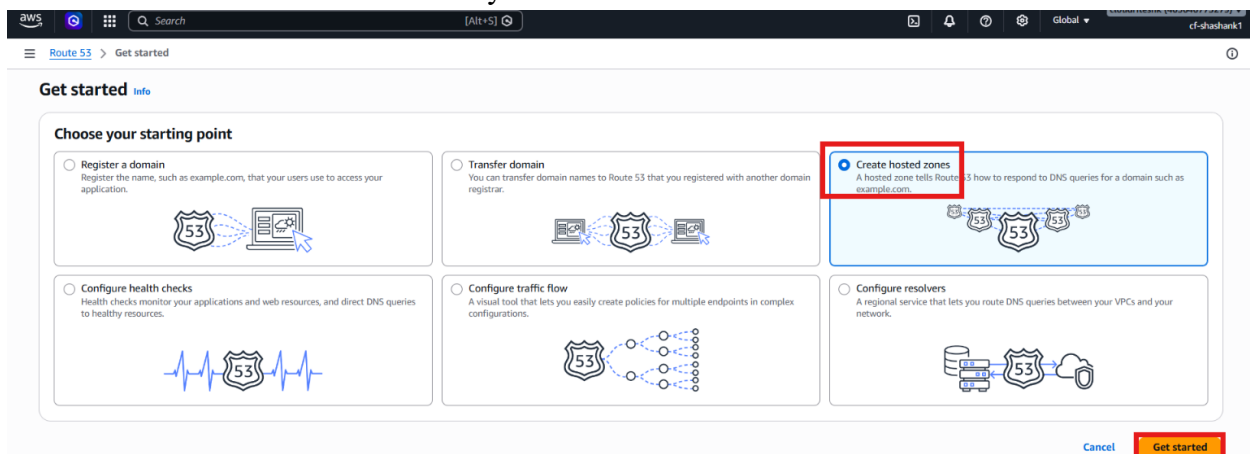
- Sign in to the AWS Management Console.
- Open the existing **Amazon S3** static website bucket.
- Verify that static website hosting is already enabled.



- Confirm that the bucket contains the website files used as the origin.
- Note that the S3 website endpoint uses **HTTP** only.

Step 2 — Register the Custom Domain

- Use a custom domain that matches the business name.
- Open **Route 53**.
- Confirm that a hosted zone already exists for the domain.



- Use the hosted zone to manage DNS records for the website.

Step 3 — Request and Validate the SSL/TLS Certificate

- Open **AWS Certificate Manager**.
- Click **Request a certificate**.

- Select **Request a public certificate**.

Request certificate

Certificate type Info
ACM certificates can be used to establish secure communications access across the internet or within an internal network. Choose the type of certificate for ACM to provide.

☒ **Request a public certificate**
Request a public SSL/TLS certificate from Amazon. By default, public certificates are trusted by browsers and operating systems.

☐ Request a private certificate
No private CAs available for issuance.

Requesting a private certificate requires the creation of a private certificate authority (CA). To create a private CA, visit [AWS Private Certificate Authority](#).

[Cancel](#) [Next](#)

- Enter the fully qualified domain name.
- Add the wildcard domain if subdomains will also be used.
- Choose **DNS validation**.
- Request the certificate.

Request public certificate

Domain names
Provide one or more domain names for your certificate.

Fully qualified domain name Info
cloudservicesdemo.in

[Add another name to this certificate](#)
You can add additional names to this certificate. For example, if you're requesting a certificate for "www.example.com", you might want to add the name "example.com" so that customers can reach your site by either name.

Allow export Info

☒ **Disable export**
Use this certificate only with integrated AWS services. The private key for this certificate will be disallowed for exporting from AWS.

☐ **Enable export**
Export this certificate and private key for use with any TLS workflow. ACM will charge your account based on the requested domains when the certificate is issued for the first time and for each renewal.

Validation method Info
Select a method for validating domain ownership.

☒ **DNS validation - recommended**
Choose this option if you are authorized to modify the DNS configuration for the domains in your certificate request.

☐ **Email validation**
Choose this option if you do not have permission or cannot obtain permission to modify the DNS configuration for the domains in your certificate request.

- Use **Create record in Route 53** to add the validation CNAME automatically.
- Wait until the certificate status changes to **Issued**.
- Verify that the certificate is created in **US East (N. Virginia)** for CloudFront use.

Step 4 — Create the CloudFront Distribution

- Open **Amazon CloudFront**.
- Click **Create distribution**.
- Select the S3 website endpoint as the origin.

- Keep the origin protocol as **HTTP**.
- Enable redirect from **HTTP to HTTPS**.
- Attach the ACM certificate from **US East (N. Virginia)**.
- Select the stronger TLS policy.

Edit settings

General

Description - optional

Alternate domain name (CNAME) - optional

Add the custom domain names that you use in URLs for the files served by this distribution.

[Add item](#)

To add a list of items, use the [bulk editor](#).

Custom SSL certificate - optional

Associate a certificate from AWS Certificate Manager. The certificate must be in the US East (N. Virginia) Region (us-east-1).

cloudservicesdemo.in (fb3af731-0e34-4576-8502-849e04adac84)

[cloudservicesdemo.in](#) [Request certificate](#)

☐ Legacy clients support - \$600/month prorated charge applies. Most customers do not need this. CloudFront allocates dedicated IP addresses at each CloudFront edge location to serve your content over HTTPS.

Security policy

The security policy determines the SSL or TLS protocol and the specific ciphers that CloudFront uses for HTTPS connections with viewers (clients).

☒ TLSv1.3_2025

☐ TLSv1.2_2025

☐ TLSv1.2_2021 (recommended)

☐ TLSv1.2_2019

☐ TLSv1.2_2018

☐ TLSv1.1_2016

- Leave the remaining options at default.
- Create the distribution and wait for deployment.

Step 5 — Map the Domain and Verify the Website

- In **Route 53**, create an alias record for the custom domain.
- Point the alias record to the CloudFront distribution.
- Add the alternate domain name in the CloudFront distribution if the custom URL shows a protocol error.
- Save the distribution changes.

Create record

Quick create record

[Switch to wizard](#)

Record 1

[Delete](#)

Record name [Info](#)

myweb .cloudservicesdemo.in

Keep blank to create a record for the root domain.

☒ Alias

Route traffic to [Info](#)

Alias to CloudFront distribution

US East (N. Virginia)

[Request for a record that is associated with another resource. Some records are global and available only in US East \(N. Virginia\).](#)

Q d164vcvljyuo2r.cloudfront.net

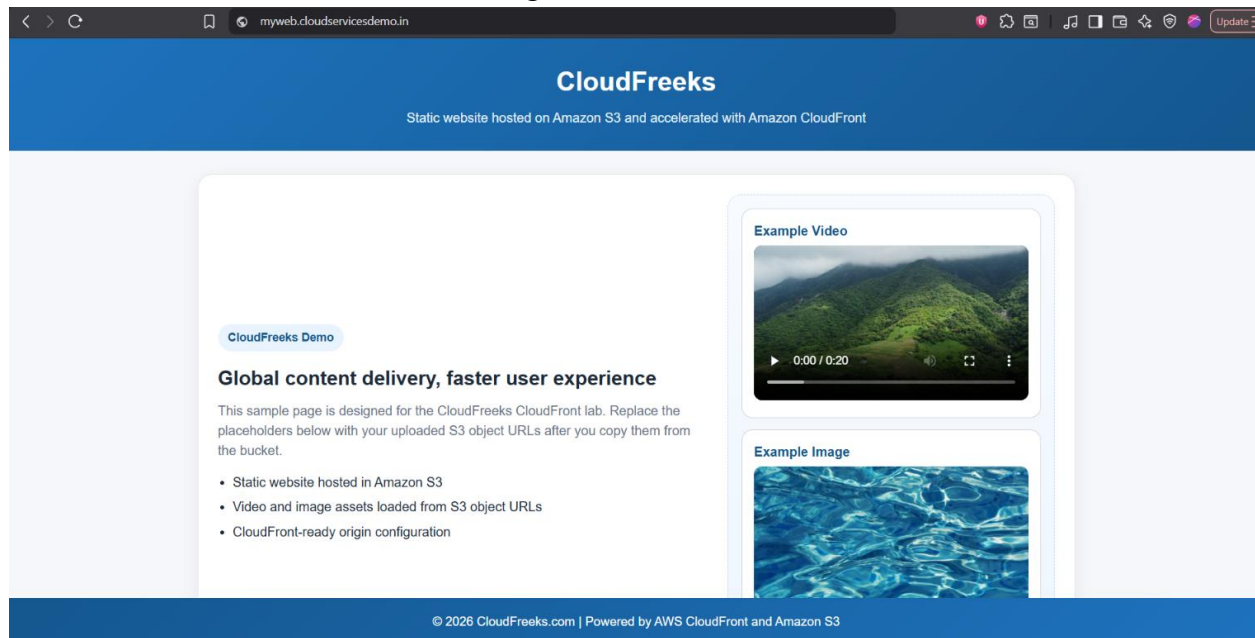
Use: "d164vcvljyuo2r.cloudfront.net"

myweb.cloudservicesdemo.in (d164vcvljyuo2r.cloudfront.net)

[Add another record](#)

[Cancel](#) [Create records](#)

- Open the custom domain using **https://**.
- Verify that the browser shows a secure connection.
- Confirm that the website loads through the branded domain.



Verification

- The S3 website was successfully delivered through **CloudFront**.
- The custom domain was mapped through **Route 53**.
- The SSL/TLS certificate was validated in **AWS Certificate Manager**.
- The website opened over **HTTPS** with a secure browser connection.
- Direct S3 access was still possible and was identified as a security issue for the next lab.